Water Through Rock Lab

BUILD · Science · 40 minutes

I can follow or direct a fixed water-passage method, record what reaches the collector and make a careful comparison of supplied samples.

Prepare before pupils arrive

Setup: Pretest all rigs for stability, sealing and a measurable result. Always prepare the fixed A/B records so learning never depends on a physical result appearing.

Safety: Local risk assessment is the authority. If a rig or sample is unmatched, unsealed, unstable, damaged, dusty or uninformative, do not improvise; use the fixed data/director route.

Equipment: Staff-built sealed rigs; pre-inspected prepared sample pieces; stable trays; 5 mL needle-free pipettes or syringes; marked collectors; 120-second timer; wipes; result sheet.

Safety: Only staff build and approve rigs. No hammering, chipping, scratching, licking, tasting, smelling or dust blowing. Keep samples on trays, contain spills, stop for leaks/damage/flaking/dust and wash hands. A director or fixed-data route is full participation.

Fallback: Use printed method and result cards. A pupil sequences the method and directs an adult scribe; no handling is required.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Test rocks for whether they let water through (permeability).

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Last lesson: what observation directly showed passage through? Last lesson: why did an edge leak fail the method check?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	permeability — how water passes through volume — amount of liquid valid — the method worked as stated Staff check the stable tray, pre-inspected matched pieces, sealed collar and clear collector. Pupils may operate only agreed safe steps or direct an adult.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Compare collected-water values only after checking that both runs were valid. A lower value means less passage was observed in the stated time, not proven permanent waterproofing.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each lab record to VALID A, VALID B, METHOD FAULT or CAREFUL COMPARISON. Check seal, volume and time on the record.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Follow or direct the fixed water-passage lab for supplied Samples A and B, or analyse the fixed equivalent records, then give a bounded comparison. Record: Method checklist, surface/underside observation, collector values with mL, valid/fault decision, bounded comparison and one method-control question for W13.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Give the comparison and explain why it does not mean waterproof forever. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: More water passed through supplied Sample B under this method.

Correct. It compares the supplied samples and stays inside the fixed time, volume and method.

Misconception repair

Repaired. The sealed route protects the meaning of collected-water evidence.

Guided checkpoint

Correct. It compares the supplied samples and stays inside the fixed time, volume and method. Choose the claim that uses the actual values without always, every or forever.

A • sealed run

VALID · SAMPLE A. Valid A record: 0.5 mL passed through under the fixed method. Match the sample code after checking the seal, volume and time.

A • surface note

VALID · SAMPLE A. Valid A observation: the record separates water on top from water collected below. The sample code and valid method place this beside the A result.

B • sealed run

VALID · SAMPLE B. Valid B record: 3.0 mL passed through under the fixed method. Use the B code and confirm the fixed method before placing.

B • underside note

VALID · SAMPLE B. Valid B observation: the water route reached the underside and collector. This observation belongs with the sample code that produced it.

Loose collar

METHOD FAULT · DO NOT COMPARE. Method fault: stop and repair or switch route; do not use this as permeability data. Did the water travel through the sample or around it?

Timer missed

METHOD FAULT · DO NOT COMPARE. Method fault: the result has no reliable shared time condition. A comparison needs the stated test time.

Bounded result

CAREFUL COMPARISON. Careful comparison: it uses the values and names the supplied samples and method. This card turns the two valid records into one bounded statement.

Time caution

CAREFUL COMPARISON. Careful comparison: the claim keeps the time and purpose limit visible. This is reasoning about how far the result can travel.

Model limitation

The lab uses a short classroom test on prepared samples. Natural variation, tiny cracks, sample thickness and sealing can affect results; the test is not a full building-performance check.

Independent evidence

Method checklist, surface/underside observation, collector values with mL, valid/fault decision, bounded comparison and one method-control question for W13.

Supported: Sequence the fixed method, place A and B result cards and complete an oral, pointed or scribed comparison. Use: A collected __ mL. B collected __ mL. More water passed through __ in this test.

Standard: Record both valid results, reject one fault record and explain the sample comparison with a time limit. Check seal, volume, time, collector and claim boundary in order.

Stretch: Evaluate how thickness, seal or natural variation could affect the comparison and propose one control to investigate in W13. Do not rewrite the observed values; improve the future method.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W11B_Water_Through_Rock_Lab_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.